2(a)(i)	gas obeys formula pV / T = constant	**M1**
	symbols V and T explained	**A1**
2(a)(ii)	mean-square-speed (of atoms / molecules)	**B1**
2(b)(i)	use of $T = 393$	**C1**
	$pV = nRT$	**C1**
	$2.4 \times 10^5 \times 6.8 \times 10^{-3} = n \times 8.31 \times 393$ and $N = n \times 6.02 \times 10^{23} = 3.0 \times 10^{23}$	**A1**
	or $pV = NkT$	**(C1)**
	$2.4 \times 10^5 \times 6.8 \times 10^{-3} = N \times 1.38 \text{ x } 10^{-23} \times 393$ hence $N = 3.0 \times 10^{23}$	**(A1)**
2(b)(ii)	volume of one atom = $4 / 3\pi r^3$	**C1**
	volume occupied $= 3.0 \times 10^{23} \times 4 / 3 \times \pi \times (3.2 \times 10^{-11})^3$ $= 4 \times 10^{-8}\ \text{m}^3$	**A1**
2(b)(iii)	assumption: volume of atoms negligible compared to volume of container / cylinder	**B1**
	$4 \times 10^{-8}\ (\text{m}^3) << 6.8 \times 10^{-3}\ (\text{m}^3)$ so yes	**B1**